

Advanced Programming

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SESSION 5 — 6 OCTOBER 2026



Would you notice?

Today's session

	MIN	WHAT
A	20	25 days of parking
B	15	You had one job · 28 of 28
C	15	Five ways to connect it to reality
D	20	Your spike
E	15	Homework 5



25 days of parking

What do you pay?

DAILY = 1000 # Ft

MONTHLY = 20000 # Ft, any 30 days in a row

```
def best_price(days):
```

You park for **25 days**. What do you pay?

25 days: two ways to pay

DAILY TICKETS

$25 \times 1,000$

25,000 Ft

MONTHLY PASS

1 pass = any 30 days in a row

20,000 Ft

Two tests, one program

```
def best_price(days):  
    return days * DAILY           # the code: forgets the pass  
  
def test_by_the_agent():  
    assert best_price(25) == 25000 # copied from running the code  
  
def test_by_the_room():  
    assert best_price(25) == 20000 # said before running
```

The agent's test passes — it copied the code. **The room's test fails** — and now you know the code is wrong.

And 45 days?

The obvious fix:

```
def best_price(days):  
    return min(days * DAILY, MONTHLY)
```

Your 25-day test is now green. **What do you pay for 45 days?**

45 days: the room's answer

1 pass (30 days) + 15 daily tickets = **35,000 Ft.** The fix said **20,000.**

```
def test_45_days():  
    assert best_price(45) == 35000      # said before running  
  
def best_price(days):  
    passes, rest = divmod(days, 30)  
    return passes * MONTHLY + min(rest * DAILY, MONTHLY)
```

Two answers from outside caught two bugs. Now both tests are green.



You had one job

You had one job



A good hob. A good ring.

Budapest, Nyugati tér, 1980



Built from both ends. In the middle, **the two halves did not meet** in height.

The legend: two different sea levels.

The record: the work was rushed — the elements were tensioned before the glue had set. Each half looked fine on its own.

You had one job



Two good handles.

Mars Climate Orbiter, 1999

What happened. One team's software sent thruster data in **pound-force seconds**. The navigation software expected **newton seconds**. Every number was off by a factor of 4.45.

Why nobody noticed. Each program was right by its own rules. The data flowed for months; small course errors were noticed, but nobody checked them against an outside number.

The outside answer. Where the spacecraft actually was. It flew too low and was lost. A \$327 million mission.

Source: NASA Mars Climate Orbiter Mishap Investigation Board, Phase I Report, 10 Nov 1999.

You had one job



A good door. Good steps.

Laufenburg, 2003



Photo: Taxiarchos228, Wikimedia Commons (CC BY-SA 3.0). Sources: Kanton Aargau press release; swissinfo.ch, "Meereshöhe ist nicht gleich Meereshöhe".

A bridge over the Rhine, Germany to Switzerland, built from both banks.

Germany measures height from the **North Sea**, Switzerland from the **Mediterranean**: 27 cm apart. They knew — and corrected it **with the wrong sign**.

The gap: **54 cm**. Caught on site, December 2003.

You had one job



Good steps. A good fence.

Ariane 5, Flight 501, 1996

What happened. 37 seconds after launch the rocket destroyed itself. A velocity value no longer fit into a 16-bit number.

Why nobody noticed. The code came from Ariane 4 and had worked for years. It was **never tested with Ariane 5's own flight path** — a faster rocket.

The outside answer. Ariane 5's real trajectory data. The inquiry asked for exactly that test.

Source: Ariane 501 Inquiry Board Report (Lions report), 19 July 1996.



28 of 28

Would your tests notice?

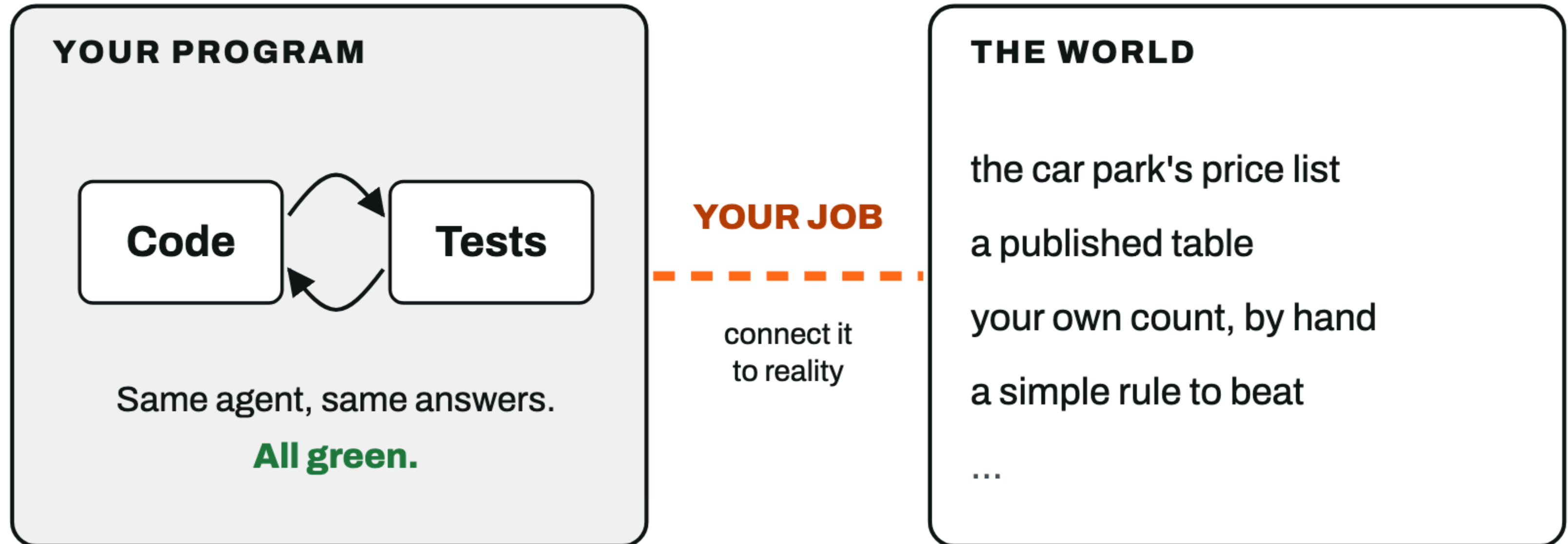
Your agents wrote **20,000 lines of tests**. They all pass.

We changed **one line** in each project, so that its main README promise became false. Then we ran your tests.

28 of 28: nothing went red.

Not a verdict on you. All 28 miss the same thing as the parking code: **a link to the real world**.

Your program agrees with itself



The one job that is yours





Five ways

Five ways to connect it to reality

WHERE THE ANSWER COMES FROM

EXAMPLE

A · You, by hand — count, label, calculate

25 days → 20,000

B · A simple rule to beat

“always the daily ticket”

C · An outside source — a table, a list, a handbook

the car park’s price list

D · Data where you planted the answer

fake respondents you made clean on purpose

E · A property that must hold

never more than 25×1000



Your spike

Your spike

Your Homework 4 number came from one of these five places.

- 1** Your project is on one of the next slides. **Which letter is your spike?**
- 2** Finish the sentence: **A test would go red if ...**

That sentence is where Homework 5 starts.

AlterEgo Finder

The promise. A HIGH_CONFIDENCE match is trustworthy: a face that is not in the dataset gets one at most about 1% of the time.

Your spike. How often does a stranger match “high”? → **99% at 500 faces.** You searched with people absent from collections of 50, 500 and 5,000 faces.

We broke it. We removed the upper threshold, so every match above “low” becomes HIGH_CONFIDENCE → **43 tests, all green.**

A test would go red if ...

Blocky

The promise. During a block, opening reddit.com shows “reddit.com is blocked until 17:00”, not an error page.

Your spike. How many blocked sites slip through? → **1 of 10**. You tried ten chosen sites in your usual and a fresh browser profile.

We broke it. We moved the app to port 8766 while the extension still calls 8765 → **87 Python and 10 Node tests, all green.**

A test would go red if ...

Budget Planner

The promise. With a food plan of 200 and 190 already spent, one more 25 of groceries shows a red warning: “Food: 15 over plan.”

Your spike. Does a real month fit six categories? → **Other 0–23%**. You sorted your own September spending into the six categories.

We broke it. We switched the over-plan check off, so the warning never appears → **8 tests, all green.**

A test would go red if ...

CAUSAL LAB

The promise. If treated and control trends already drift apart before treatment, the app flags it instead of calling it fine.

Your spike. How often do the checks cry wolf? → **57.6 of 100**. You ran the checks 200 times on a flawless simulated world, then with each bias switched on.

We broke it. We set the parallel-trends threshold so it can never warn → **12 tests, all green**.

A test would go red if ...

Central European air traffic

The promise. A record with no position lands in the rejects table with a reason; the run still commits the rest.

Your spike. How much time did the collector cover? → **6.8% of hours.** You counted fully covered hours over two weeks: 23 of 337.

We broke it. We changed “OR” to “AND” in the missing-position rule, so a half-missing record vanishes → **no test ran, so nothing could notice.**

A test would go red if ...

ChatGPT Ads Industry Dashboard

The promise. The box plot shows real differences between industries, bigger than ordinary noise.

Your spike. Industry gap bigger than weekly noise? → **143% vs 56%**. You compared the real campaign's week-to-week swing with the gap between industries.

We broke it. We gave every synthetic account the same click rate and cost per click. The gap shrank to 11%, the decision flipped → **36 tests, all green.**

A test would go red if ...

Data Jobs & Skill-Gap Finder for Early-Career Professionals

The promise. Excel and R rank as distinctive skills, because “R&D” and “excel in” are not counted.

Your spike. Are the skill labels true? → **98.8% precision.** You hand-labelled 50 postings, the same 50 the patterns were tuned on.

We broke it. We restored the old R and Excel patterns that match “R&D” and “excel in” → **13 tests, all green.**

A test would go red if ...

deriva

The promise. Between two survey rounds, the program separates cosmetic description changes from real ones, the way a person would.

Your spike. Does it read rounds it has never seen? → **3 of 3.** You loaded the 2022, 2023 and 2024 dictionaries without changing the code.

We broke it. The rule stopped ignoring one accent mark; the demo split moved from 9/13 to 8/14 → **53 tests, all green.**

A test would go red if ...

Detective game with LLM-driven NPCs

The promise. A suspect never says anything that is not in their own approved answer, so they cannot reveal what they do not know.

Your spike. Does the real model path work? → **30 of 30** · **2.2 s.** You made 30 calls to the real model and checked each reply and its response time.

We broke it. We handed the model every suspect's secrets in the prompt → **11 tests, all green.**

A test would go red if ...

European bank financial data loader

The promise. For a chosen bank and period, the package returns the reported figures, ready for metrics such as RWA density.

Your spike. Do the two sources agree? → **RWA density 44.5%**. You checked the figures against the bank's own published reports.

We broke it. We pointed risk-weighted assets at the wrong reporting item; RWA density would go above 100% → **8 tests, all green.**

A test would go red if ...

Explainable Fraud Detection

The promise. The report gives a shortlist of the transactions worth a closer look, instead of scanning everything by hand.

Your spike. How many flags are real fraud? → **68 of 500.** You checked the flags against the fraud labels of a public dataset.

We broke it. We used the median instead of the average amount; flags on your slice jumped from 500 to 907 → **58 tests, all green.**

A test would go red if ...

Financial Planner

The promise. Moves between your own accounts drop out of spending and income, and nothing else does, so the monthly summary is true.

Your spike. Does it categorise new shops? → **64.8% new · 95.5% seen.** You labelled 551 expense rows by hand and compared them with the program's categories.

We broke it. We let a transfer match on any date, not only the same day → **154 tests, all green.**

A test would go red if ...

Market Research Platform

The promise. The report names each question's leading answers with percentages, and every figure can be checked against the data.

Your spike. How tightly can a cell be estimated? → **6–34 points wide.** You computed the possible range for 35 cells from the aggregate export.

We broke it. We reversed the sort order, so the smallest answer is called “leading” (19.1% instead of 59.0%) → **there are no tests to notice.**

A test would go red if ...

MarketLens

The promise. Every expired forecast is scored against the actual closing price on its due date.

Your spike. Better than “always up”? → **52.3% vs 54.8%**.
You replayed 501 days and compared the model’s hit rate with a rule that always says “Up”.

We broke it. We made scoring take the wrong day’s price (the day before, or the very first day) → **26 tests, all green.**

A test would go red if ...

Menstrual Cycle Phase Calculator and Targeted Nutrition Planner

The promise. For today's phase it prints the nutrients your evidence review pairs with it, each with its evidence rating.

Your spike. Is there a source for each claim? → **7 of 12 (2 strong)**. You checked the 12 phase–nutrient pairs against medical databases and official guidance.

We broke it. We swapped the nutrient lists of two phases in the rules file → **57 tests, all green.**

A test would go red if ...

Ms. Match

The promise. On a rainy day (50% or more), the suggested shoes are rain-proof whenever you own such a pair.

Your spike. How many garments fit? → **84 (63 your own)**.
You added ten of your own photos and measured how much browser storage each one takes.

We broke it. We removed the line that keeps only rain-proof shoes on rainy days → **there are no tests to notice.**

A test would go red if ...

Photo Duplicate Finder

The promise. At the default setting, bursts and resized copies land in one group, while photos that are merely similar stay apart.

Your spike. Which threshold is safe on real photos? → **1, not 5.** You ran thresholds 0 to 12 on about 300 of your own photos and compared group sizes.

We broke it. We changed the default threshold from 8 to 0, then to 25 → **36 tests, all green.**

A test would go red if ...

PixWars

The promise. Two laptops on the room's wifi can play together at 20 updates a second without lag or lost packets.

Your spike. Is the wifi fast enough at 20 Hz? → **p99 61 ms**.
You timed round trips from a laptop to a phone on your home wifi.

We broke it. We made the echo program never reply, a total network outage → **114 tests, all green**, and the report still shows 0% loss.

A test would go red if ...

plant_care

The promise. Each card counts down to when the plant really needs watering.

Your spike. Do the watering days match a horticultural source? → **0 of 12 give days.** You checked 12 plants against RHS pages; none states a day count.

We broke it. We set the cactus to be watered every day and the Boston fern every 60 days → **122 tests, all green.**

A test would go red if ...

Process Swimlane Generator

The promise. Every arrow runs from a task's box to its next task's box, without crossing any other box.

Your spike. Does it draw real processes? → **4 of 5**. You ran five real, anonymised workplace processes and judged each diagram yourself.

We broke it. We bent every arrow back into its own box, so none reaches the next task → **28 tests green, exactly as before** (the 4 red ones are Windows-only).

A test would go red if ...

RC Atlas

The promise. Each role sees only its own workflow buttons: field submits, MEAL validates, the manager approves.

Your spike. How many steps survive before saving fails? → **10 of 10.** You ran a fixed 10-step workflow on a fresh install and counted the steps that worked.

We broke it. We switched off the condition that shows the MEAL officer the Validate button → **201 tests, all green.**

A test would go red if ...

RepoScout

The promise. “6 active in the last 3 months” counts the repos that had a commit in the last 3 months.

Your spike. How wrong is “last updated”? → **5 of 50 misfiled.** You compared GitHub’s push date with each repo’s real last commit.

We broke it. We made every repo count as “active”, whatever its dates → **47 tests, all green.**

A test would go red if ...

Route Tariff and Capacity Fee Calculator

The promise. For a route and period, the screen shows the cheapest way to book it, with the saving.

Your spike. How much does exact cover overpay? → **worst 135.8%**. You compared the exact-cover plan with the cheapest over-booking plan in 1,050 cases.

We broke it. We made the yearly product cost almost nothing; the “cheapest” route did not change → **53 route tests, all green.**

A test would go red if ...

sentmine

The promise. If a word is in your known list, its -s, -ed and -ing forms do not become cards either.

Your spike. How many cards are really new words? → **4 of 103**. You marked by hand every card from a 1,220-word Guardian article.

We broke it. We limited the -e rule to short words, so “promise” no longer covers “promising” → **225 tests, all green.**

A test would go red if ...

Stardew Valley Auto-Gifter

The promise. A villager who already got two gifts this week, or one today, gets no more automatic gifts.

Your spike. Does the chest keep its colour? → **3 of 3 openings.** You logged the chest colour at every opening, also after save and reload.

We broke it. We could not run it, but the gift-limit code is not even in the test build → **the 11 tests cannot go red.**

A test would go red if ...

SurveyGuard: Numeric Careless Responding Detector

The promise. A respondent who gives the same answer to every item is marked suspicious, ready to be excluded.

Your spike. How many clean respondents get flagged? → **25.4% · 71.9%**. You ran the three flags on 1,000 public personality-test responses.

We broke it. We made the score mark nobody as suspicious, ever → **15 tests, all green.**

A test would go red if ...

Toto Estimator

The promise. On matches it has not seen, the predictor gets more results right than always tipping “1”.

Your spike. Better than “always 1”? → **40% vs 46%**. On a held-out season of 380 real English league matches, your predictor against always “1”.

We broke it. We made it predict “1” for every match → **38 tests, all green.**

A test would go red if ...

Volleyball Statistics Tracker

The promise. Paste a real CEV match URL, and the team totals come out the same as CEV's own.

Your spike. Do the totals match the CEV sheet? → **22 of 24 cells.** You compared your totals, cell by cell, with CEV's printed Team Totals row.

We broke it. We swapped two columns in the importer: attack errors and blocked attacks → **20 tests, all green.**

A test would go red if ...

Homework 5 — Make it usable

- Use your program **once more for real** this week, on your own input. Expected before, got after — in the log.
- The first mismatch becomes **a test that can fail**: show it red, then green.
- If it can't be used yet: remove **the smallest thing** that stops you, then use it.
- Branch `hw5-usable`, pull request, merge commit — as before.

Due Saturday 11 October, 20:00.